THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

MATH2241
Introduction to Mathematical Analysis

Tutorial 3 — Solutions

Question 1.

Recall

$$x_n = \cos\left(\frac{n\pi}{2}\right) + \frac{1}{n}.$$

(a) We have

$$x_{4k} = \cos(2k\pi) + \frac{1}{4k} = 1 + \frac{1}{4k} \longrightarrow 1,$$

and

$$x_{4k+2} = \cos((2k+1)\pi) + \frac{1}{4k+2} = -1 + \frac{1}{4k+2} \longrightarrow -1.$$

Also,

$$x_{2k+1} = \cos\left(\frac{(2k+1)\pi}{2}\right) + \frac{1}{2k+1} = \frac{1}{2k+1} \longrightarrow 0.$$

(b) Since (x_{4k}) and (x_{4k+2}) are subsequences of (x_n) with different limits, (x_n) cannot converge.

(c) The set of subsequential limits is

$$\boxed{\{-1, 0, 1\}}.$$

Indeed, each n belongs to one of the four residue classes modulo 4:

$$4k, \qquad 4k+1, \qquad 4k+2, \qquad 4k+3.$$

The corresponding subsequences converge respectively to

$$1, \qquad 0, \qquad -1, \qquad 0.$$

Any infinite subsequence contains infinitely many indices from at least one of these four classes. Hence, if that subsequence converges, its limit must be one of $-1, 0, 1$.

Intuition. The term $1/n$ disappears in the limit, so the long-run behaviour is determined by the periodic values of $\cos(n\pi/2)$.

Question 2.

Suppose, for contradiction, that

$$x_n \not\longrightarrow L.$$

Then there exists $\varepsilon > 0$ such that for every $N \in \mathbb{N}$, there exists $n > N$ satisfying

$$|x_n - L| \geq \varepsilon.$$

Hence we can choose a subsequence (x_{n_k}) such that

$$|x_{n_k} - L| \geq \varepsilon \qquad \text{for every } k.$$

Since (x_n) is bounded, (x_{n_k}) is bounded. By the Bolzano–Weierstrass Theorem, (x_{n_k}) has a convergent subsequence, say

$$x_{n_{k_j}} \longrightarrow M.$$

By hypothesis, every convergent subsequence of (x_n) converges to L, so $M = L$. Thus

$$x_{n_{k_j}} \longrightarrow L.$$

Therefore, for sufficiently large j,

$$|x_{n_{k_j}} - L| < \varepsilon,$$

contradicting

$$|x_{n_{k_j}} - L| \geq \varepsilon.$$

Hence

$$\boxed{x_n \longrightarrow L}.$$

Intuition. If the sequence stayed a fixed positive distance away from L infinitely often, those terms would themselves contain a convergent subsequence with a limit different from L.

Question 3.

(a) Let $m > n \geq 2$. By the triangle inequality,

$$|x_m - x_n| \leq \sum_{k=n}^{m-1} |x_{k+1} - x_k| \leq \sum_{k=n}^{m-1} \frac{1}{k^2}.$$

For $k \geq 2$,

$$\frac{1}{k^2} \leq \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}.$$

Hence

$$|x_m - x_n| \leq \sum_{k=n}^{m-1} \left(\frac{1}{k-1} - \frac{1}{k} \right) = \frac{1}{n-1} - \frac{1}{m-1} < \frac{1}{n-1}.$$

Given $\varepsilon > 0$, choose N such that

$$\frac{1}{N-1} < \varepsilon.$$

Then, whenever $m > n > N$,

$$|x_m - x_n| < \varepsilon.$$

Therefore (x_n) is Cauchy.

(b) Let

$$x_n = \sqrt{n}.$$

Then

$$x_{n+1} - x_n = \sqrt{n+1} - \sqrt{n} = \frac{1}{\sqrt{n+1} + \sqrt{n}} \longrightarrow 0.$$

However, (x_n) is not Cauchy. Take $\varepsilon = 1$. For any N, choose $n > N$ and set $m = 4n$. Then

$$|x_m - x_n| = |\sqrt{4n} - \sqrt{n}| = \sqrt{n} > 1.$$

Thus the Cauchy condition fails.

Intuition. Small consecutive steps do not guarantee that the *total* movement over many steps is small. In part (a), the step sizes have a summable tail; in part (b), they do not.

Question 4.

(a) Since, for $n \geq 1$,

$$n + 4 \leq 5n \qquad \text{and} \qquad n^3 + 2n + 1 \geq n^3,$$

we have

$$0 < \frac{n+4}{n^3+2n+1} \leq \frac{5n}{n^3} = \frac{5}{n^2}.$$

Since

$$\sum_{n=1}^{\infty} \frac{5}{n^2}$$

converges, the Comparison Test gives

$$\boxed{\sum_{n=1}^{\infty} \frac{n+4}{n^3+2n+1} \text{ converges.}}$$

(b) Let

$$a_n = \frac{\sqrt{n^2+1}}{n+2}.$$

Then

$$a_n = \frac{\sqrt{1+\frac{1}{n^2}}}{1+\frac{2}{n}} \longrightarrow 1.$$

Since $a_n \not\to 0$, the n-th term test implies

$$\boxed{\sum_{n=1}^{\infty} \frac{\sqrt{n^2+1}}{n+2} \text{ diverges.}}$$

(c) Since

$$n^2 + \sqrt{n} \geq n^2,$$

we have

$$0 < \frac{1}{n^2+\sqrt{n}} \leq \frac{1}{n^2}.$$

Since $\sum 1/n^2$ converges, the Comparison Test gives

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n^2+\sqrt{n}} \text{ converges.}}$$

Question 5.

Consider

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^{2/3}}.$$

(a) Let

$$b_n = \frac{1}{n^{2/3}}.$$

Then (b_n) is decreasing and

$$b_n \longrightarrow 0.$$

Hence the Alternating Series Test implies that

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^{2/3}}$$

converges.

However,

$$\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{n^{2/3}} \right| = \sum_{n=1}^{\infty} \frac{1}{n^{2/3}}$$

is a p-series with $p = 2/3 < 1$, so it diverges.

Therefore the original series is

$$\boxed{\text{conditionally convergent.}}$$

(b) Let

$$s_n = \sum_{k=1}^{n} a_k.$$

The m-th partial sum of the grouped series is

$$\begin{aligned} t_m &= (a_1 + a_2) + (a_3 + a_4) + \cdots + (a_{2m-1} + a_{2m}) \\ &= \sum_{k=1}^{2m} a_k \\ &= s_{2m}. \end{aligned}$$

Since the original series converges, there exists S such that

$$s_n \longrightarrow S.$$

Therefore its subsequence also converges to S:

$$t_m = s_{2m} \longrightarrow S.$$

Hence the grouped series converges to the same value as the original series.

(c) Grouping consecutive terms does not change their order. It merely places parentheses around consecutive finite blocks. Thus the partial sums of the grouped series form a subsequence of the original partial sums.

A genuine rearrangement, in contrast, changes the order in which the terms appear. For a conditionally convergent series, changing this order can change the behaviour of the partial sums.

Intuition. Grouping keeps the original order intact; rearrangement does not.

Question 6.

Recall

$$a_n = \begin{cases} 3 \cdot 2^{-n}, & n \text{ odd}, \\ 2^{-n}, & n \text{ even}. \end{cases}$$

(a) If n is odd, then $n+1$ is even, so

$$\frac{a_{n+1}}{a_n} = \frac{2^{-(n+1)}}{3 \cdot 2^{-n}} = \frac{1}{6}.$$

If n is even, then $n+1$ is odd, so

$$\frac{a_{n+1}}{a_n} = \frac{3 \cdot 2^{-(n+1)}}{2^{-n}} = \frac{3}{2}.$$

Thus

$$\frac{a_{n+1}}{a_n} = \begin{cases} \frac{1}{6}, & n \text{ odd}, \\ \frac{3}{2}, & n \text{ even}, \end{cases}$$

so

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n}$$

does not exist.

Therefore the Ratio Test, in the form stated in the lecture, is

$$\boxed{\text{inconclusive.}}$$

(b) If n is even,

$$\sqrt[n]{a_n} = \sqrt[n]{2^{-n}} = \frac{1}{2}.$$

If n is odd,

$$\sqrt[n]{a_n} = \sqrt[n]{3 \cdot 2^{-n}} = \frac{\sqrt[n]{3}}{2} \longrightarrow \frac{1}{2}.$$

Hence

$$\lim_{n\to\infty} \sqrt[n]{a_n} = \frac{1}{2} < 1.$$

By the Root Test,

$$\boxed{\sum_{n=1}^{\infty} a_n \text{ converges.}}$$

(c) This example shows that when a convergence test is *inconclusive*, it means only that the hypotheses of that test do not allow us to decide the question.

It does *not* mean that the series diverges. Here the Ratio Test is inconclusive, while the Root Test proves convergence.

Intuition. A convergence test is a sufficient tool, not a definition of convergence. Failure of a test is not failure of the series.